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May 2015

MBR3035PT - MBR3060PT
30 A Schottky Barrier Rectifiers

Features

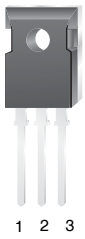
- Low Power Loss, High Efficiency
- High Surge Capacity
- Metal Silicon Junction, Majority Carrier Conduction
- High Current Capacity, Low Forward Voltage Drop
- Guard Ring for Over-Voltage Protection (OVP)

Applications

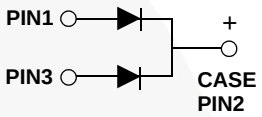
- Low-Voltage
- High-Frequency Inverters
- Free Wheeling
- Polarity Protection

Description

This center-tap Schottky rectifier is optimal for secondary rectification and free-wheeling applications for high-efficiency DC-DC convertor design, which features very low forward voltage drop and low leakage current.



TO-3P/TO-247AD



Ordering Information

Part Number	Marking	Package	Packing Method
MBR3035PT	MBR3035PT	TO-247 3L	Rail
MBR3045PT	MBR3045PT		
MBR3050PT	MBR3050PT		
MBR3060PT	MBR3060PT		

Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value				Unit
		MBR 3035PT	MBR 3045PT	MBR 3050PT	MBR 3060PT	
V_{RRM}	Maximum Repetitive Reverse Voltage	35	45	50	60	V
$I_{F(AV)}$	Average Rectified Forward Current .375-inch Lead Length	30				A
I_{FSM}	Non-Repetitive Peak Forward Surge Current: 8.3 ms Single Half-Sine Wave	200				A
T_{STG}	Storage Temperature Range	-65 to +175				°C
T_J	Operating Junction Temperature Range	-65 to +150				°C

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Thermal Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value	Unit
P_D	Power Dissipation	3.0	W
$R_{\theta JC}$	Thermal Resistance, Junction to Case	1.4	°C/W

Electrical Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter		Value				Unit
			MBR 3035PT	MBR 3045PT	MBR 3050PT	MBR 3060PT	
V_F	Maximum Forward Voltage, per Leg	I_F = 20 A, T_C = 25°C			0.75		V
		I_F = 20 A, T_C = 125°C	0.60		0.65		
		I_F = 30 A, T_C = 25°C	0.76				
		I_F = 30 A, T_C = 125°C	0.72				
I_R	Maximum Reverse Current at Rated V_{RRM}, per Leg	T_A = 25°C	0.1		5.0		mA
		T_A = 125°C	60.0		100.0		
I_{RRM}	Peak Repetitive Reverse Surge Current, per Leg 2.0 µs Pulse Width, f = 1.0 kHz		1.0		0.5		A

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Typical Performance Characteristics

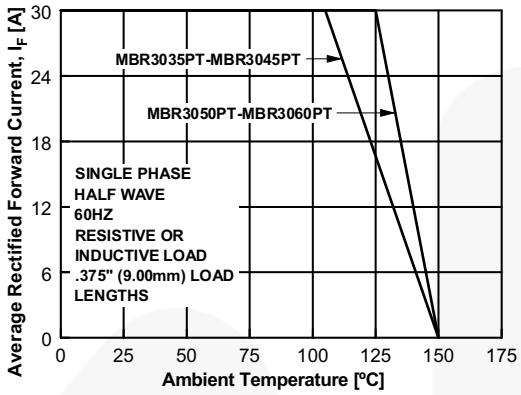


Figure 1. Forward Current Derating Curve

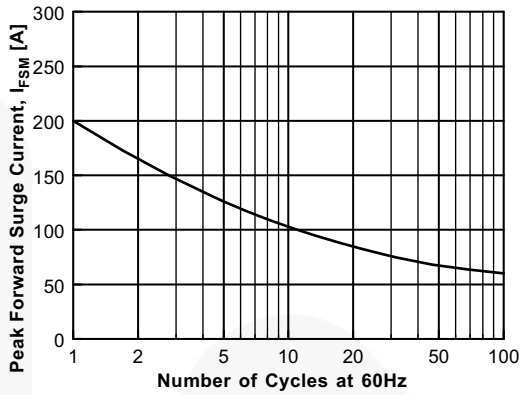


Figure 2. Non-Repetitive Surge Current

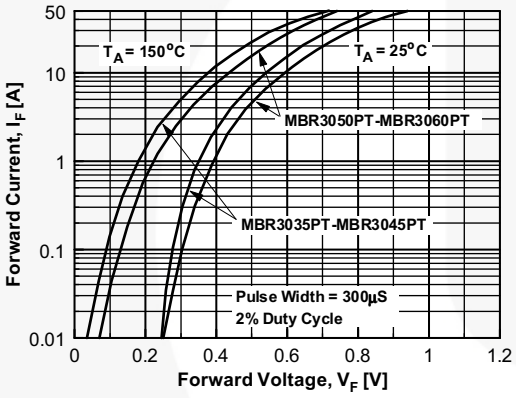


Figure 3. Forward Voltage Characteristics

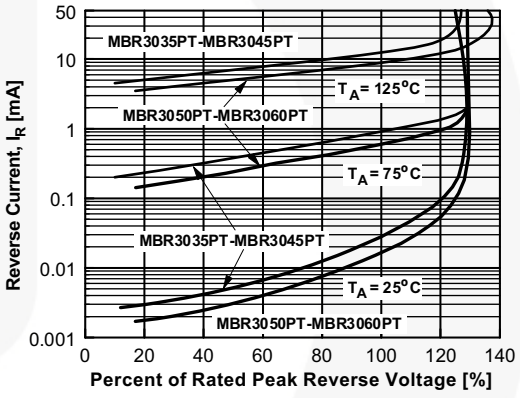


Figure 4. Reverse Current vs. Reverse Voltage

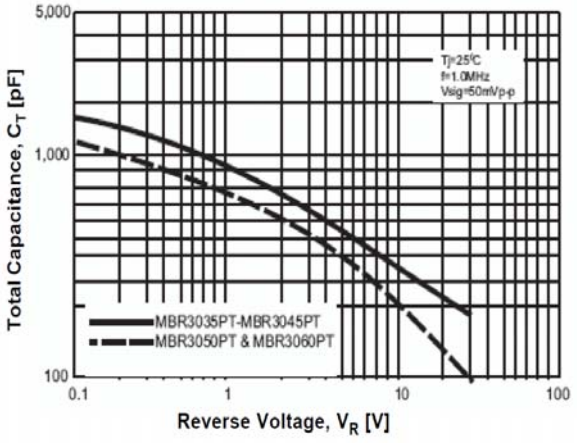


Figure 5. Total Capacitance

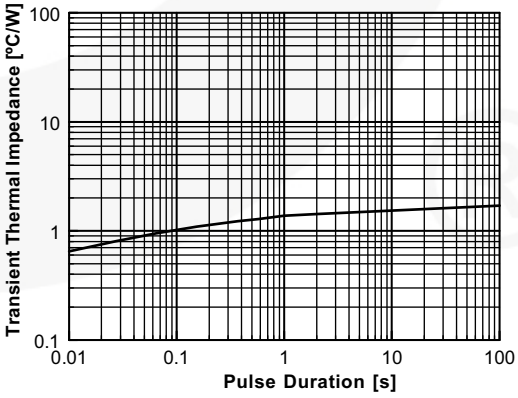
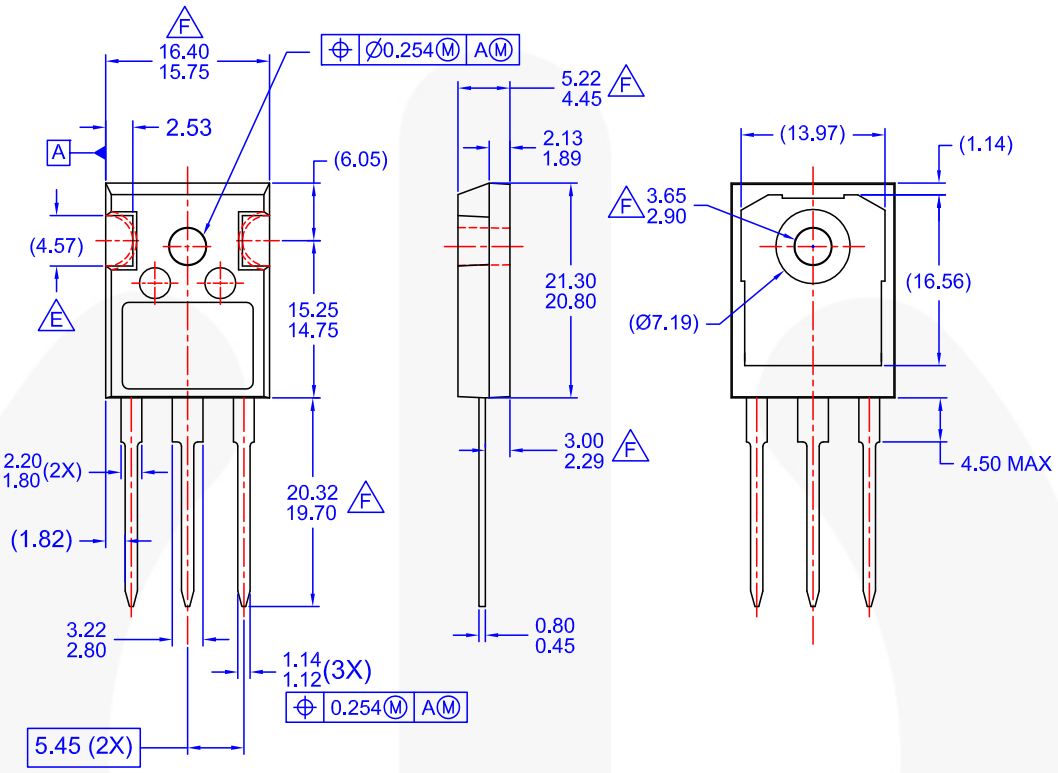


Figure 6. Thermal Impedance Characteristics

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Physical Dimensions



NOTES: UNLESS OTHERWISE SPECIFIED

A. PACKAGE REFERENCE: JEDEC TO-247, ISSUE "E", VARIATION AD

B. DIMENSIONS ARE EXCLUSIVE OF BURRS, MOLD FLASH, AND TIE BAR EXTRUSIONS.

C. ALL DIMENSIONS ARE IN MILLIMETERS.

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E. DOES NOT COMPLY JEDEC STANDARD VALUE.

F. NOTCH MAY BE SQUARE

G. DRAWING FILENAME: MKT-TO247E03_REV02

Figure 7. TO-247, MOLDED, 3 LEADS, JEDEC OPTION AD

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